

Exponential Expressions*Independent Set 3.03* **ANSWER KEY***Six questions, on your own, in silence. Show the step that does the work.*

1. Find the value. $\frac{4^2 \cdot 4^0}{4^1}$

$$4^1 = 4$$

4^0 is 1; then $4^2 \div 4^1$.

2. Find the value. $(2^0 \cdot 2^3)^2$

$$(2^3)^2 = 2^6 = 64$$

3. Write as one power. $\left(\frac{5^8}{5^5}\right)^3$

$$5^9$$

Inside first gives 5^3 ; then $(5^3)^3$.

4. Find the value. $0.2^2 \cdot (0.2^1)^2$

$$0.2^4 = 0.0016$$

5. Find the value. $\frac{(-3)^3 \cdot (-3)^0}{(-3)^2}$

$$(-3)^1 = -3$$

6. Show that $\left(\frac{10^4}{10^2}\right)^2$ and $\frac{10^8}{10^4}$ have the same value.

Both are $10^4 = 10,000$.

Two plans, one value — the point of the lesson.